SQL- ASSIGNMENT-4 (SQL SELECT Queries Assignment)

Assignments Questions:-

Example 1: List all the records in the student table

- select * from students;
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	1	John	Doe	2005-01-15	M	10Math	90
2	2	Jane	Smith	1989-02-22	F	10Sci	95
3	3	Alice	Johnson	2005-03-18	F	10His	93
4	4	Sophia	Brown	2004-12-20	F	9His	81
5	5	Robert	Davis	1989-04-05	M	10Bio	97
6	6	Diana	Miller	2005-05-25	M	10Math	86
7	7	Emma	Wilson	2005-06-12	F	10Math	92
8	8	Krish	Moore	2004-07-10	M	9Bio	85
9	9	Krish	Taylor	2005-08-09	F	10Sci	88
10	10	Kiya	Anderson	2004-09-14	M	10Math	83
11	11	Vikas	Nelson	2005-10-10	M	10Math	89
12	12	Sally	Allen	2006-03-14	F	9Bio	94
13	13	Linda	Sandra	2005-12-22	F	11His	96
14	14	Tho...	Nelson	2005-10-10	M	10Math	89
15	18	Vish	Shah	1900-01-01	F	10Math	80
16	19	Vishwa	Shah	1900-01-01	F	10Math	80
17	20	Vishl	Shah	1900-01-01	M	10Math	80

Example 2: List the name surname and class of the student in the student table

- select name,surname,class from students;

- Output:

	name	surname	class
1	John	Doe	10Math
2	Jane	Smith	10Sci
3	Alice	Johnson	10His
4	Sophia	Brown	9His
5	Robert	Davis	10Bio
6	Diana	Miller	10Math
7	Emma	Wilson	10Math
8	Krish	Moore	9Bio
9	Krish	Taylor	10Sci
10	Kiya	Anderson	10Math
11	Vikas	Nelson	10Math
12	Sally	Allen	9Bio
13	Linda	Sandra	11His
14	Thomas	Nelson	10Math
15	Vish	Shah	10Math
16	Vishwa	Shah	10Math
17	Vishl	Shah	10Math

Example 3: List the gender Female (F) records in the student table

- select * from students where gender ='F';
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	2	Jane	Smith	1989-02-22	F	10Sci	95
2	3	Alice	Johnson	2005-03-18	F	10His	93
3	4	Sophia	Brown	2004-12-20	F	9His	81
4	7	Emma	Wilson	2005-06-12	F	10Math	92
5	9	Krish	Taylor	2005-08-09	F	10Sci	88
6	12	Sally	Allen	2006-03-14	F	9Bio	94
7	13	Linda	Sandra	2005-12-22	F	11His	96
8	18	Vish	Shah	1900-01-01	F	10Math	80
9	19	Vishwa	Shah	1900-01-01	F	10Math	80

Example 4 : List the names of each class in the way of being seen once in the student table

- select distinct class from students;
- Output:

	class
1	10Bio
2	10His
3	10Math
4	10Sci
5	11His
6	9Bio
7	9His

Example 5: List the students with Female gender and the class 10Math in the student table

- select * from students where gender='F' and class='10Math';
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	7	Emma	Wilson	2005-06-12	F	10Math	92
2	18	Vish	Shah	1900-01-01	F	10Math	80
3	19	Vishwa	Shah	1900-01-01	F	10Math	80

Example 6: List the names, surnames and classes of the students in the class 10Math or 10Sci in the student table

- select name,surname,class from students where class='10Math' or class='10Sci';
- Output:

	name	surname	class
1	John	Doe	10Math
2	Jane	Smith	10Sci
3	Diana	Miller	10Math
4	Emma	Wilson	10Math
5	Krish	Taylor	10Sci
6	Kiya	Anderson	10Math
7	Vikas	Nelson	10Math
8	Thomas	Nelson	10Math
9	Vish	Shah	10Math
10	Vishwa	Shah	10Math
11	Vishl	Shah	10Math

Example 7: List the students name surname and school number in the student table

- select name,surname,studentId from students;
- Output:

	name	surname	studentId
1	John	Doe	1
2	Jane	Smith	2
3	Alice	Johnson	3
4	Sophia	Brown	4
5	Robert	Davis	5
6	Diana	Miller	6
7	Emma	Wilson	7
8	Krish	Moore	8
9	Krish	Taylor	9
10	Kiya	Anderson	10
11	Vikas	Nelson	11
12	Sally	Allen	12
13	Linda	Sandra	13
14	Thomas	Nelson	14
15	Vish	Shah	18
16	Vishwa	Shah	19
17	Vishl	Shah	20

Example 8: List the students name and surname by combining them as name surname in the student table

- select name + ' ' + surname as "Full Name" from students;
- Output:

	Full Name
1	John Doe
2	Jane Smith
3	Alice Johnson
4	Sophia Brown
5	Robert Davis
6	Diana Miller
7	Emma Wilson
8	Krish Moore
9	Krish Taylor
10	Kiya Anderson
11	Vikas Nelson
12	Sally Allen
13	Linda Sandra
14	Thomas Nelson
15	Vish Shah
16	Vishwa Shah
17	Vishl Shah

Example 9: List the students with the names starting with “A” letter in the student table

- select * from students where name like 'A%';
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	3	Alice	Johnson	2005-03-18	F	10His	93

Example 10: List the book names and pages count with number of pages between 50 and 200 in the book table

- select name,pagecount from books where pagecount between 50 and 200;
- Output:

	name	pagecount
1	The Great Gatsby	180

Example 11: List the students with names Emma Sophia and Robert in the student table

- select * from students where name IN ('Emma','Sophia','Robert');
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	4	Sophia	Brown	2004-12-20	F	9His	81
2	5	Robert	Davis	1989-04-05	M	10Bio	97
3	7	Emma	Wilson	2005-06-12	F	10Math	92

Example 12: List the students with names starting with A D and K in the student table

- select * from students where name like 'A%' or name like 'D%' or name like 'K%';
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	3	Alice	Johnson	2005-03-18	F	10His	93
2	6	Diana	Miller	2005-05-25	M	10Math	86
3	8	Krish	Moore	2004-07-10	M	9Bio	85
4	9	Krish	Taylor	2005-08-09	F	10Sci	88
5	10	Kiya	Anderson	2004-09-14	M	10Math	83

Example 13: List the names surnames classes and genders of males in 9Math or females in 9His in the student table

- select name,surname,class,gender from students where (gender ='M' and class ='9Math') or (gender ='F' and class ='9His');
- Output:

	name	surname	class	gender
1	Sophia	Brown	9His	F

Example 14: List the males whose classes are 10Math or 10Bio

- select * from students where gender = 'M' and class IN ('10Math','10Bio');
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	1	John	Doe	2005-01-15	M	10Math	90
2	5	Robert	Davis	1989-04-05	M	10Bio	97
3	6	Diana	Miller	2005-05-25	M	10Math	86
4	10	Kiya	Anderson	2004-09-14	M	10Math	83
5	11	Vikas	Nelson	2005-10-10	M	10Math	89
6	14	Thomas	Nelson	2005-10-10	M	10Math	89
7	20	Vishl	Shah	1900-01-01	M	10Math	80

Example 15: List the students with birth year 1989 in the student table

- select * from students where YEAR(birthdate) = 1989;
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	2	Jane	Smith	1989-02-22	F	10Sci	95
2	5	Robert	Davis	1989-04-05	M	10Bio	97

Example 16: List the female students with student numbers between 30 and 50

- SELECT * FROM students WHERE gender='F' AND studentId BETWEEN 30 AND 50;
- Output:

studentId	name	surname	birthdate	gender	class	point

Example 17: List the students according to their names

- select * from students order by name;
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	3	Alice	Johnson	2005-03-18	F	10His	93
2	6	Diana	Miller	2005-05-25	M	10Math	86
3	7	Emma	Wilson	2005-06-12	F	10Math	92
4	2	Jane	Smith	1989-02-22	F	10Sci	95
5	1	John	Doe	2005-01-15	M	10Math	90
6	10	Kiya	Anderson	2004-09-14	M	10Math	83
7	8	Krish	Moore	2004-07-10	M	9Bio	85
8	9	Krish	Taylor	2005-08-09	F	10Sci	88
9	13	Linda	Sandra	2005-12-22	F	11His	96
10	5	Robert	Davis	1989-04-05	M	10Bio	97
11	12	Sally	Allen	2006-03-14	F	9Bio	94
12	4	Sophia	Brown	2004-12-20	F	9His	81
13	14	Thomas	Nelson	2005-10-10	M	10Math	89
14	11	Vikas	Nelson	2005-10-10	M	10Math	89
15	18	Vish	Shah	1900-01-01	F	10Math	80
16	20	Vishl	Shah	1900-01-01	M	10Math	80
17	19	Vishwa	Shah	1900-01-01	F	10Math	80

Example 18: List the students by names for those with same names. List them by their surnames

- SELECT * FROM students ORDER BY name, surname;
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	3	Alice	Johnson	2005-03-18	F	10His	93
2	6	Diana	Miller	2005-05-25	M	10Math	86
3	7	Emma	Wilson	2005-06-12	F	10Math	92
4	2	Jane	Smith	1989-02-22	F	10Sci	95
5	1	John	Doe	2005-01-15	M	10Math	90
6	10	Kiya	Anderson	2004-09-14	M	10Math	83
7	8	Krish	Moore	2004-07-10	M	9Bio	85
8	9	Krish	Taylor	2005-08-09	F	10Sci	88
9	13	Linda	Sandra	2005-12-22	F	11His	96
10	5	Robert	Davis	1989-04-05	M	10Bio	97
11	12	Sally	Allen	2006-03-14	F	9Bio	94
12	4	Sophia	Brown	2004-12-20	F	9His	81
13	14	Thomas	Nelson	2005-10-10	M	10Math	89
14	11	Vikas	Nelson	2005-10-10	M	10Math	89
15	18	Vish	Shah	1900-01-01	F	10Math	80
16	20	Vishl	Shah	1900-01-01	M	10Math	80
17	19	Vishwa	Shah	1900-01-01	F	10Math	80

Example 19: List the students in 10Math by decreasing school numbers

- select * from students where class = '10Math' order by studentId desc;
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	20	Vishl	Shah	1900-01-01	M	10Math	80
2	19	Vishwa	Shah	1900-01-01	F	10Math	80
3	18	Vish	Shah	1900-01-01	F	10Math	80
4	14	Thomas	Nelson	2005-10-10	M	10Math	89
5	11	Vikas	Nelson	2005-10-10	M	10Math	89
6	10	Kiya	Anderson	2004-09-14	M	10Math	83
7	7	Emma	Wilson	2005-06-12	F	10Math	92
8	6	Diana	Miller	2005-05-25	M	10Math	86
9	1	John	Doe	2005-01-15	M	10Math	90

Example 20: List the first 10 records in the student chart

- SELECT top 10 * FROM students;

- Output:

	studentId	name	surname	birthdate	gender	class	point
1	1	John	Doe	2005-01-15	M	10Math	90
2	2	Jane	Smith	1989-02-22	F	10Sci	95
3	3	Alice	Johnson	2005-03-18	F	10His	93
4	4	Sophia	Brown	2004-12-20	F	9His	81
5	5	Robert	Davis	1989-04-05	M	10Bio	97
6	6	Diana	Miller	2005-05-25	M	10Math	86
7	7	Emma	Wilson	2005-06-12	F	10Math	92
8	8	Krish	Moore	2004-07-10	M	9Bio	85
9	9	Krish	Taylor	2005-08-09	F	10Sci	88
10	10	Kiya	Anderson	2004-09-14	M	10Math	83

Example 21: List the first 10 records name surname and date of birth information in the student table

- select top 10 name,surname,birthdate FROM students;
- Output:

	name	surname	birthdate
1	John	Doe	2005-01-15
2	Jane	Smith	1989-02-22
3	Alice	Johnson	2005-03-18
4	Sophia	Brown	2004-12-20
5	Robert	Davis	1989-04-05
6	Diana	Miller	2005-05-25
7	Emma	Wilson	2005-06-12
8	Krish	Moore	2004-07-10
9	Krish	Taylor	2005-08-09
10	Kiya	Anderson	2004-09-14

Example 22: List the book with the most page number

- select top 1 * from books order by pagecount desc;
- Output:

	bookId	name	pagecount	point	authorId	typeId
1	8	Annihilation	1225	92	8	7

Example 23: List the youngest student in the student table

- select top 1 * from students order by birthdate;

- Output:

	studentId	name	surname	birthdate	gender	class	point
1	18	Vish	Shah	1900-01-01	F	10Math	80

Example 24: List the oldest student in the 10Math class

- SELECT TOP 1 * FROM students WHERE class = '10Math' ORDER BY birthdate ASC;
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	18	Vish	Shah	1900-01-01	F	10Math	80

Example 25: List the books with the second letter N

- select * from books where name like '_N%';
- Output:

	bookId	name	pagecount	point	authorId	typeId
1	2	Infinity War	281	90	2	2
2	8	Annihilation	1225	92	8	7

Example 26: List the students by grouping according to their classes

- select class,count(*) as total_students from students group by class;
- Output:

	class	total_students
1	10Bio	1
2	10His	1
3	10Math	9
4	10Sci	2
5	11His	1
6	9Bio	2
7	9His	1

Example 27: List the students to be different in each questioning randomly

- select * from students order by rand();
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	1	John	Doe	2005-01-15	M	10Math	90
2	2	Jane	Smith	1989-02-22	F	10Sci	95
3	3	Alice	Johnson	2005-03-18	F	10His	93
4	4	Sophia	Brown	2004-12-20	F	9His	81
5	5	Robert	Davis	1989-04-05	M	10Bio	97
6	6	Diana	Miller	2005-05-25	M	10Math	86
7	7	Emma	Wilson	2005-06-12	F	10Math	92
8	8	Krish	Moore	2004-07-10	M	9Bio	85
9	9	Krish	Taylor	2005-08-09	F	10Sci	88
10	10	Kiya	Anderson	2004-09-14	M	10Math	83
11	11	Vikas	Nelson	2005-10-10	M	10Math	89
12	12	Sally	Allen	2006-03-14	F	9Bio	94
13	13	Linda	Sandra	2005-12-22	F	11His	96
14	14	Thomas	Nelson	2005-10-10	M	10Math	89
15	18	Vish	Shah	1900-01-01	F	10Math	80
16	19	Vishwa	Shah	1900-01-01	F	10Math	80
17	20	Vishl	Shah	1900-01-01	M	10Math	80

Example 28: Pick a random student from student table

- select top 1 * from students order by rand();
- Output:

	studentId	name	surname	birthdate	gender	class	point
1	1	John	Doe	2005-01-15	M	10Math	90

Example 29: Bring some random student's name , surname and number from class 10Math

- SELECT name, surname, studentId FROM students WHERE class = '10Math' ORDER BY RAND();
- Output:

	name	surname	studentId
1	John	Doe	1
2	Diana	Miller	6
3	Emma	Wilson	7
4	Kiya	Anderson	10
5	Vikas	Nelson	11
6	Thomas	Nelson	14
7	Vish	Shah	18
8	Vishwa	Shah	19
9	Vishl	Shah	20

Example 30: Add the writer named Smith Allen to the authors table

- INSERT INTO authors (authorId, name, surname) VALUES (12,'Smith', 'Allen');

Example 31: Add the genre of biography to the genre table

- INSERT INTO types (typeId, name) VALUES (11, 'Biography');

Example 32: Add 10Math Class male named Thomas Nelson , 9Bio class female named Sally Allen and 11His Class female named Linda Sandra in one question

- INSERT INTO students (studentId, name, surname, birthdate, gender, class, point) VALUES
  (14, 'Thomas', 'Nelson', '2005-10-10', 'M', '10Math', 84),
  (12, 'Sally', 'Allen', '2006-03-14', 'F', '9Bio', 89),
  (13, 'Linda', 'Sandra', '2005-12-22', 'F', '11His', 91);

Example 34: Add students with student numbers between 10 and 30 as authors

- INSERT INTO authors (name, surname) SELECT name, surname FROM students
  WHERE studentId BETWEEN 10 AND 30;

Example 35: Add the writer named Cindy Brown and make him write his writer number (Note: The last increased rate in automatic enhancing is hold in @@IDENTITY factor)

- INSERT INTO authors (authorId,name, surname) VALUES (30,'Riya', 'Brown');
  SELECT @@IDENTITY AS LastInsertedAuthorId;

Example 36: Change the class of the student whose school number is 3 from 10Bio to 10His

- UPDATE students SET class='10His' WHERE studentId=3 AND class='10Bio';

Example 37: Transfer all the students in 9Math Class to 10Math Clas

- update students set class='10Math' where class='9Math';

Example 38: Increase all of the students' score by 5 points

- update students set point = point + 5;

Example 39: Delete the author #25

- delete from authors where authorId = 25;

Example 40: List the students whose birth dates are null

- select * from students where birthdate is NULL;
- Output:

studentId	name	surname	birthdate	gender	class	point

Example 41: List the name ,surname and the dates of received books of the student

- SELECT s.name, s.surname, b.takenDate, b.broughtDate FROM students s JOIN borrows b ON s.studentId = b.studentId;
- Output:

	name	surname	takenDate	broughtDate
1	John	Doe	2024-01-10	2024-01-20
2	Jane	Smith	2024-01-12	2024-01-22
3	Alice	Johnson	2024-01-15	2024-01-25
4	Sophia	Brown	2024-01-18	2024-01-28
5	Robert	Davis	2024-01-20	2024-01-30
6	Diana	Miller	2024-01-22	2024-02-01
7	Emma	Wilson	2024-01-24	2024-02-03
8	Krish	Moore	2024-01-26	2024-02-05
9	Krish	Taylor	2024-01-28	2024-02-07
10	Kiya	Anderson	2024-01-30	2024-02-09